Assignment 5

2026-06-18

```r
library(stats)
library(cluster)
library(factoextra)
```

```
## Loading required package: ggplot2

## Welcome to factoextra!

## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practica
```

```r
cereals<- read.csv("~/Desktop/Cereals.csv")

na.omit(cereals)
```

```
##                                    name mfr type calories protein fat sodium
## 1                             100%_Bran   N    C       70       4   1    130
## 2                     100%_Natural_Bran   Q    C      120       3   5     15
## 3                              All-Bran   K    C       70       4   1    260
## 4             All-Bran_with_Extra_Fiber   K    C       50       4   0    140
## 6               Apple_Cinnamon_Cheerios   G    C      110       2   2    180
## 7                           Apple_Jacks   K    C      110       2   0    125
## 8                               Basic_4   G    C      130       3   2    210
## 9                             Bran_Chex   R    C       90       2   1    200
## 10                          Bran_Flakes   P    C       90       3   0    210
## 11                         Cap'n'Crunch   Q    C      120       1   2    220
## 12                             Cheerios   G    C      110       6   2    290
## 13                Cinnamon_Toast_Crunch   G    C      120       1   3    210
## 14                             Clusters   G    C      110       3   2    140
## 15                          Cocoa_Puffs   G    C      110       1   1    180
## 16                            Corn_Chex   R    C      110       2   0    280
## 17                          Corn_Flakes   K    C      100       2   0    290
## 18                            Corn_Pops   K    C      110       1   0     90
## 19                        Count_Chocula   G    C      110       1   1    180
## 20                   Cracklin'_Oat_Bran   K    C      110       3   3    140
## 22                              Crispix   K    C      110       2   0    220
## 23               Crispy_Wheat_&_Raisins   G    C      100       2   1    140
## 24                          Double_Chex   R    C      100       2   0    190
## 25                          Froot_Loops   K    C      110       2   1    125
## 26                       Frosted_Flakes   K    C      110       1   0    200
## 27                  Frosted_Mini-Wheats   K    C      100       3   0      0
## 28 Fruit_&_Fibre_Dates,_Walnuts,_and_Oats   P    C      120       3   2    160
## 29                        Fruitful_Bran   K    C      120       3   0    240
```

```
## 30                       Fruity_Pebbles   P    C      110       1   1    135
## 31                         Golden_Crisp   P    C      100       2   0     45
## 32                       Golden_Grahams   G    C      110       1   1    280
## 33                    Grape_Nuts_Flakes   P    C      100       3   1    140
## 34                           Grape-Nuts   P    C      110       3   0    170
## 35                   Great_Grains_Pecan   P    C      120       3   3     75
## 36                     Honey_Graham_Ohs   Q    C      120       1   2    220
## 37                   Honey_Nut_Cheerios   G    C      110       3   1    250
## 38                           Honey-comb   P    C      110       1   0    180
## 39          Just_Right_Crunchy__Nuggets   K    C      110       2   1    170
## 40               Just_Right_Fruit_&_Nut   K    C      140       3   1    170
## 41                                  Kix   G    C      110       2   1    260
## 42                                 Life   Q    C      100       4   2    150
## 43                        Lucky_Charms   G    C      110       2   1    180
## 44                                Maypo   A    H      100       4   1      0
## 45      Muesli_Raisins,_Dates,_&_Almonds   R    C      150       4   3     95
## 46    Muesli_Raisins,_Peaches,_&_Pecans   R    C      150       4   3    150
## 47                  Mueslix_Crispy_Blend   K    C      160       3   2    150
## 48                  Multi-Grain_Cheerios   G    C      100       2   1    220
## 49                     Nut&Honey_Crunch   K    C      120       2   1    190
## 50           Nutri-Grain_Almond-Raisin   K    C      140       3   2    220
## 51                   Nutri-grain_Wheat   K    C       90       3   0    170
## 52                 Oatmeal_Raisin_Crisp   G    C      130       3   2    170
## 53              Post_Nat._Raisin_Bran   P    C      120       3   1    200
## 54                          Product_19   K    C      100       3   0    320
## 55                         Puffed_Rice   Q    C       50       1   0      0
## 56                        Puffed_Wheat   Q    C       50       2   0      0
## 57                   Quaker_Oat_Squares   Q    C      100       4   1    135
## 59                         Raisin_Bran   K    C      120       3   1    210
## 60                     Raisin_Nut_Bran   G    C      100       3   2    140
## 61                      Raisin_Squares   K    C       90       2   0      0
## 62                           Rice_Chex   R    C      110       1   0    240
## 63                       Rice_Krispies   K    C      110       2   0    290
## 64                      Shredded_Wheat   N    C       80       2   0      0
## 65               Shredded_Wheat_'n'Bran   N    C       90       3   0      0
## 66           Shredded_Wheat_spoon_size   N    C       90       3   0      0
## 67                              Smacks   K    C      110       2   1     70
## 68                           Special_K   K    C      110       6   0    230
## 69              Strawberry_Fruit_Wheats   N    C       90       2   0     15
## 70                    Total_Corn_Flakes   G    C      110       2   1    200
## 71                    Total_Raisin_Bran   G    C      140       3   1    190
## 72                    Total_Whole_Grain   G    C      100       3   1    200
## 73                             Triples   G    C      110       2   1    250
## 74                                Trix   G    C      110       1   1    140
## 75                          Wheat_Chex   R    C      100       3   1    230
## 76                            Wheaties   G    C      100       3   1    200
## 77                  Wheaties_Honey_Gold   G    C      110       2   1    200
##    fiber carbo sugars potass vitamins shelf weight cups   rating
## 1   10.0   5.0      6    280       25     3   1.00 0.33 68.40297
## 2    2.0   8.0      8    135        0     3   1.00 1.00 33.98368
## 3    9.0   7.0      5    320       25     3   1.00 0.33 59.42551
## 4   14.0   8.0      0    330       25     3   1.00 0.50 93.70491
## 6    1.5  10.5     10     70       25     1   1.00 0.75 29.50954
## 7    1.0  11.0     14     30       25     2   1.00 1.00 33.17409
```

```
## 8    2.0  18.0       8    100        25      3    1.33 0.75 37.03856
## 9    4.0  15.0       6    125        25      1    1.00 0.67 49.12025
## 10   5.0  13.0       5    190        25      3    1.00 0.67 53.31381
## 11   0.0  12.0      12     35        25      2    1.00 0.75 18.04285
## 12   2.0  17.0       1    105        25      1    1.00 1.25 50.76500
## 13   0.0  13.0       9     45        25      2    1.00 0.75 19.82357
## 14   2.0  13.0       7    105        25      3    1.00 0.50 40.40021
## 15   0.0  12.0      13     55        25      2    1.00 1.00 22.73645
## 16   0.0  22.0       3     25        25      1    1.00 1.00 41.44502
## 17   1.0  21.0       2     35        25      1    1.00 1.00 45.86332
## 18   1.0  13.0      12     20        25      2    1.00 1.00 35.78279
## 19   0.0  12.0      13     65        25      2    1.00 1.00 22.39651
## 20   4.0  10.0       7    160        25      3    1.00 0.50 40.44877
## 22   1.0  21.0       3     30        25      3    1.00 1.00 46.89564
## 23   2.0  11.0      10    120        25      3    1.00 0.75 36.17620
## 24   1.0  18.0       5     80        25      3    1.00 0.75 44.33086
## 25   1.0  11.0      13     30        25      2    1.00 1.00 32.20758
## 26   1.0  14.0      11     25        25      1    1.00 0.75 31.43597
## 27   3.0  14.0       7    100        25      2    1.00 0.80 58.34514
## 28   5.0  12.0      10    200        25      3    1.25 0.67 40.91705
## 29   5.0  14.0      12    190        25      3    1.33 0.67 41.01549
## 30   0.0  13.0      12     25        25      2    1.00 0.75 28.02576
## 31   0.0  11.0      15     40        25      1    1.00 0.88 35.25244
## 32   0.0  15.0       9     45        25      2    1.00 0.75 23.80404
## 33   3.0  15.0       5     85        25      3    1.00 0.88 52.07690
## 34   3.0  17.0       3     90        25      3    1.00 0.25 53.37101
## 35   3.0  13.0       4    100        25      3    1.00 0.33 45.81172
## 36   1.0  12.0      11     45        25      2    1.00 1.00 21.87129
## 37   1.5  11.5      10     90        25      1    1.00 0.75 31.07222
## 38   0.0  14.0      11     35        25      1    1.00 1.33 28.74241
## 39   1.0  17.0       6     60       100      3    1.00 1.00 36.52368
## 40   2.0  20.0       9     95       100      3    1.30 0.75 36.47151
## 41   0.0  21.0       3     40        25      2    1.00 1.50 39.24111
## 42   2.0  12.0       6     95        25      2    1.00 0.67 45.32807
## 43   0.0  12.0      12     55        25      2    1.00 1.00 26.73451
## 44   0.0  16.0       3     95        25      2    1.00 1.00 54.85092
## 45   3.0  16.0      11    170        25      3    1.00 1.00 37.13686
## 46   3.0  16.0      11    170        25      3    1.00 1.00 34.13976
## 47   3.0  17.0      13    160        25      3    1.50 0.67 30.31335
## 48   2.0  15.0       6     90        25      1    1.00 1.00 40.10596
## 49   0.0  15.0       9     40        25      2    1.00 0.67 29.92429
## 50   3.0  21.0       7    130        25      3    1.33 0.67 40.69232
## 51   3.0  18.0       2     90        25      3    1.00 1.00 59.64284
## 52   1.5  13.5      10    120        25      3    1.25 0.50 30.45084
## 53   6.0  11.0      14    260        25      3    1.33 0.67 37.84059
## 54   1.0  20.0       3     45       100      3    1.00 1.00 41.50354
## 55   0.0  13.0       0     15         0      3    0.50 1.00 60.75611
## 56   1.0  10.0       0     50         0      3    0.50 1.00 63.00565
## 57   2.0  14.0       6    110        25      3    1.00 0.50 49.51187
## 59   5.0  14.0      12    240        25      2    1.33 0.75 39.25920
## 60   2.5  10.5       8    140        25      3    1.00 0.50 39.70340
## 61   2.0  15.0       6    110        25      3    1.00 0.50 55.33314
## 62   0.0  23.0       2     30        25      1    1.00 1.13 41.99893
## 63   0.0  22.0       3     35        25      1    1.00 1.00 40.56016
```

```
## 64   3.0  16.0      0     95        0     1   0.83 1.00 68.23588
## 65   4.0  19.0      0    140        0     1   1.00 0.67 74.47295
## 66   3.0  20.0      0    120        0     1   1.00 0.67 72.80179
## 67   1.0   9.0     15     40       25     2   1.00 0.75 31.23005
## 68   1.0  16.0      3     55       25     1   1.00 1.00 53.13132
## 69   3.0  15.0      5     90       25     2   1.00 1.00 59.36399
## 70   0.0  21.0      3     35      100     3   1.00 1.00 38.83975
## 71   4.0  15.0     14    230      100     3   1.50 1.00 28.59278
## 72   3.0  16.0      3    110      100     3   1.00 1.00 46.65884
## 73   0.0  21.0      3     60       25     3   1.00 0.75 39.10617
## 74   0.0  13.0     12     25       25     2   1.00 1.00 27.75330
## 75   3.0  17.0      3    115       25     1   1.00 0.67 49.78744
## 76   3.0  17.0      3    110       25     1   1.00 1.00 51.59219
## 77   1.0  16.0      8     60       25     1   1.00 0.75 36.18756
```

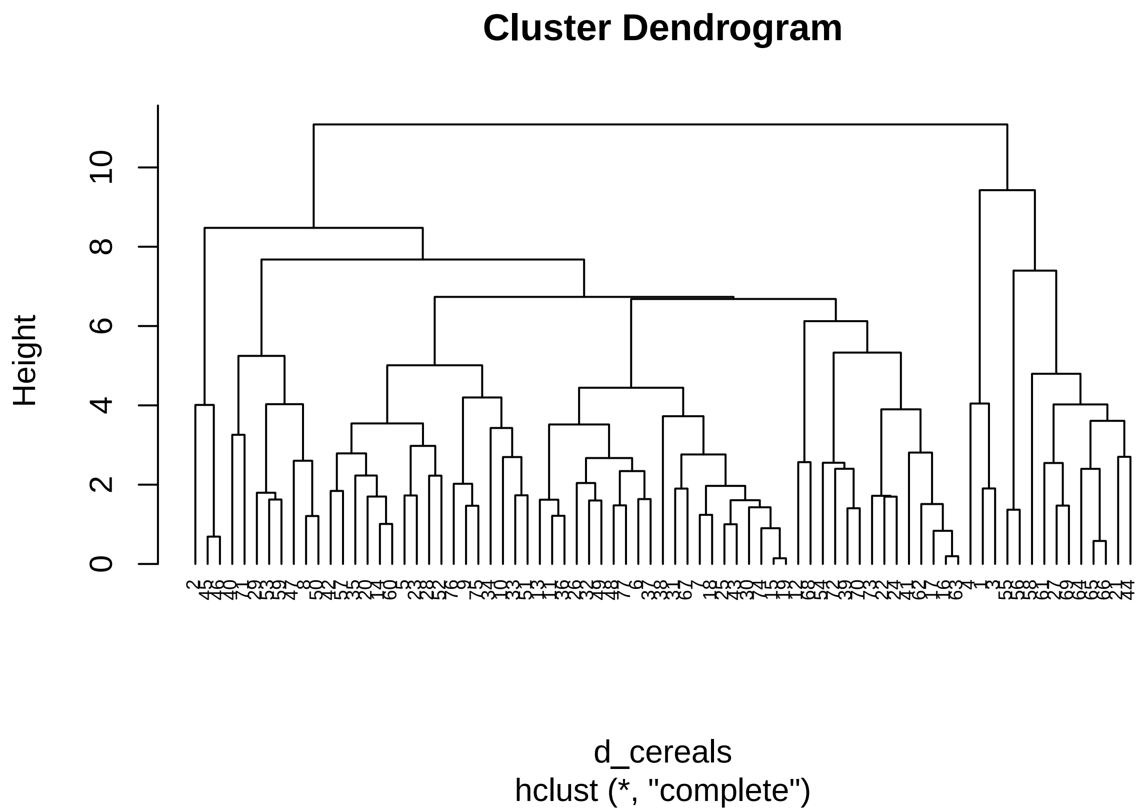
```r
head(cereals)
```

```
##                        name mfr type calories protein fat sodium fiber carbo
## 1                 100%_Bran   N    C       70       4   1    130  10.0   5.0
## 2         100%_Natural_Bran   Q    C      120       3   5     15   2.0   8.0
## 3                  All-Bran   K    C       70       4   1    260   9.0   7.0
## 4 All-Bran_with_Extra_Fiber   K    C       50       4   0    140  14.0   8.0
## 5            Almond_Delight   R    C      110       2   2    200   1.0  14.0
## 6   Apple_Cinnamon_Cheerios   G    C      110       2   2    180   1.5  10.5
##   sugars potass vitamins shelf weight cups   rating
## 1      6    280       25     3      1 0.33 68.40297
## 2      8    135        0     3      1 1.00 33.98368
## 3      5    320       25     3      1 0.33 59.42551
## 4      0    330       25     3      1 0.50 93.70491
## 5      8     NA       25     3      1 0.75 34.38484
## 6     10     70       25     1      1 0.75 29.50954
```

```r
cereals_num<- cereals[, sapply(cereals, is.numeric)]

cereals_norm<- scale(cereals_num)
```

Part 1

```r
d_cereals<- dist(cereals_norm, method = "euclidean")

hc1<- hclust(d_cereals, method = "complete")

plot(hc1, cex = 0.6, hang = -1)
```




```r
hc_single<- agnes(cereals_norm, method = "single")

hc_complete<- agnes(cereals_norm, method = "complete")

hc_average<- agnes(cereals_norm, method = "average")

hc_ward<- agnes(cereals_norm, method = "ward")
```

```r
print(hc_single$ac)
```

```
## [1] 0.6023689
```

```r
print(hc_complete$ac)
```

```
## [1] 0.8387548
```

```r
print(hc_average$ac)
```

```
## [1] 0.7768921
```

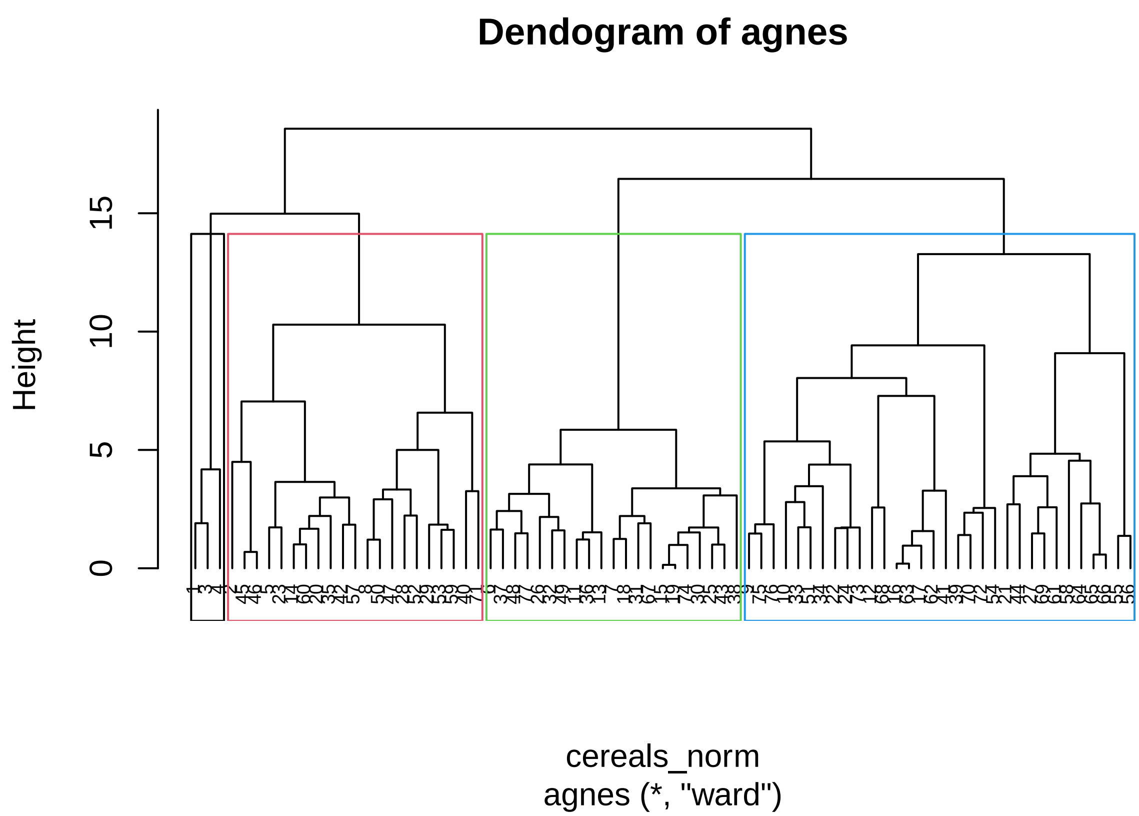
```r
print(hc_ward$ac)
```

```
## [1] 0.9029485
```

Since Ward clustering gave the highest AC (0.9046042), that is the best method.

Part 2

```r
pltree(hc_ward, cex = 0.6, hang = -1, main = "Dendogram of agnes")

rect.hclust(as.hclust(hc_ward), k = 4, border = 1:4)
```



Following the approach demonstrated in the lecture, the dendrogram was cut at a height that produced four distinct clusters. The four-cluster solution appears reasonable because the groups remain relatively separate before being merged at larger distances.

Part 3

```r
cluster_4<- cutree(as.hclust(hc_ward), k = 4)

cereals$Cluster<- as.factor(cluster_4)

table(cereals$Cluster)
```

```
##
##  1  2  3  4
##  3 21 21 32
```

```
cluster_summary<- aggregate(cereals_num, by = list(Cluster = cereals$Cluster), FUN = mean)

cluster_summary
```

```
##   Cluster  calories  protein       fat   sodium      fiber     carbo    sugars
## 1       1  63.33333 4.000000 0.6666667 176.6667 11.0000000  6.666667  3.666667
## 2       2 123.33333 3.095238 1.9523810 157.1429  3.0000000 13.952381  9.285714
## 3       3 110.95238 1.523810 1.0000000 172.3810  0.5714286 12.619048 11.285714
## 4       4  97.50000 2.718750 0.4375000 151.4062  1.8031250        NA        NA
##      potass vitamins    shelf    weight      cups   rating
## 1 310.00000 25.00000 3.000000 1.0000000 0.3866667 73.84446
## 2        NA 30.95238 2.904762 1.1642857 0.6952381 38.07700
## 3  45.95238 25.00000 1.666667 1.0000000 0.8871429 28.84825
## 4        NA 28.90625 2.031250 0.9634375 0.9009375 51.82174
```

The four clusters show different nutritional profiles based on the normalized cereal measurements. To interpret the structure, I compared the mean values of the cereal variables within each cluster.

The four clusters exhibit distinct nutritional profiles. Cluster 1 contains cereals with the lowest average calories and the highest protein content, while also achieving the highest average consumer rating. Cluster 2 consists of higher-calorie cereals with relatively high fat and sugar content and lower ratings. Cluster 3 contains cereals with the lowest protein content, relatively high sugar levels, and the lowest average ratings. Cluster 4 represents a moderate group with average calorie levels, low fat content, lower sugar levels than Clusters 2 and 3, and higher ratings than those groups. These differences suggest that the clustering successfully separated cereals based on meaningful nutritional characteristics.

```
set.seed(123)

A_index<- sample(1:nrow(cereals_num), size = round(0.5 * nrow(cereals_num)))

A<- cereals_num[A_index,]

B<- cereals_num[-A_index,]
```

```
hc_A<- agnes(A, method = "ward")

cluster_A<- cutree(as.hclust(hc_A), k = 4)

table(cluster_A)
```

```
## cluster_A
##  1  2  3  4
## 14 13  7  4
```

```
centroids_A<- aggregate(A, by = list(cluster_A), FUN = mean)

centroids_A
```

```
##   Group.1 calories  protein       fat   sodium     fiber    carbo    sugars
## 1       1 110.7143 1.571429 0.7857143 157.5000 0.4285714 13.14286 11.285714
## 2       2 112.3077 3.076923 1.3076923 173.0769 3.1153846 14.57692  7.769231
## 3       3 108.5714 2.285714 0.7142857 271.4286 0.4285714 20.00000  3.285714
```

```
## 4       4  92.5000 3.000000 0.2500000   0.0000 2.5000000 16.25000  2.500000
##      potass vitamins    shelf   weight      cups   rating
## 1  41.42857 35.71429 1.928571 1.000000 0.9557143 29.91588
## 2 143.07692 30.76923 2.615385 1.108462 0.7130769 42.92262
## 3  50.00000 25.00000 1.571429 1.000000 1.0542857 40.19125
## 4 107.50000 12.50000 1.500000 0.957500 0.8675000 63.97622
```

```r
centroid_values <- centroids_A[, -1]

centroid_values
```

```
##   calories  protein       fat   sodium     fiber    carbo    sugars    potass
## 1 110.7143 1.571429 0.7857143 157.5000 0.4285714 13.14286 11.285714  41.42857
## 2 112.3077 3.076923 1.3076923 173.0769 3.1153846 14.57692  7.769231 143.07692
## 3 108.5714 2.285714 0.7142857 271.4286 0.4285714 20.00000  3.285714  50.00000
## 4  92.5000 3.000000 0.2500000   0.0000 2.5000000 16.25000  2.500000 107.50000
##   vitamins    shelf   weight      cups   rating
## 1 35.71429 1.928571 1.000000 0.9557143 29.91588
## 2 30.76923 2.615385 1.108462 0.7130769 42.92262
## 3 25.00000 1.571429 1.000000 1.0542857 40.19125
## 4 12.50000 1.500000 0.957500 0.8675000 63.97622
```

```r
B_assigned <- apply(B, 1, function(record) {
  distances <- apply(centroid_values, 1, function(center) {
    sqrt(sum((record - center)^2))
  })

  which.min(distances)
})

B_assigned
```

```
## $`1`
## [1] 2
##
## $`2`
## [1] 4
##
## $`3`
## [1] 2
##
## $`4`
## [1] 2
##
## $`5`
## integer(0)
##
## $`6`
## [1] 1
##
## $`8`
## [1] 2
##
```

```
## $`11`
## [1] 3
##
## $`13`
## [1] 1
##
## $`16`
## [1] 3
##
## $`18`
## [1] 1
##
## $`20`
## [1] 2
##
## $`21`
## integer(0)
##
## $`22`
## [1] 3
##
## $`24`
## [1] 1
##
## $`28`
## [1] 2
##
## $`29`
## [1] 2
##
## $`30`
## [1] 1
##
## $`33`
## [1] 1
##
## $`34`
## [1] 2
##
## $`35`
## [1] 4
##
## $`37`
## [1] 3
##
## $`40`
## [1] 2
##
## $`46`
## [1] 2
##
## $`47`
## [1] 2
##
```

```
## $`48`
## [1] 3
##
## $`54`
## [1] 3
##
## $`55`
## [1] 4
##
## $`56`
## [1] 4
##
## $`58`
## integer(0)
##
## $`59`
## [1] 2
##
## $`60`
## [1] 2
##
## $`61`
## [1] 4
##
## $`66`
## [1] 4
##
## $`68`
## [1] 3
##
## $`69`
## [1] 4
##
## $`72`
## [1] 2
##
## $`76`
## [1] 2
##
## $`77`
## [1] 1
```

```r
full_clusters <- cutree(as.hclust(hc_ward), k = 4)

B_full_clusters <- full_clusters[-A_index]
```

```r
unlist(B_assigned)
```

```
##  1  2  3  4  6  8 11 13 16 18 20 22 24 28 29 30 33 34 35 37 40 46 47 48 54 55
##  2  4  2  2  1  2  3  1  3  1  2  3  1  2  2  1  1  2  4  3  2  2  2  3  3  4
## 56 59 60 61 66 68 69 72 76 77
##  4  2  2  4  4  3  4  2  2  1
```

```r
# Use the same normalized data for the full clustering and partitions
set.seed(123)

A_index <- sample(1:nrow(cereals_norm),
                  size = round(0.5 * nrow(cereals_norm)))

A <- cereals_norm[A_index, ]
B <- cereals_norm[-A_index, ]

# Cluster partition A
hc_A <- agnes(A, method = "ward")
cluster_A <- cutree(as.hclust(hc_A), k = 4)

# Get centroids from A
centroids_A <- aggregate(A,
                         by = list(Cluster = cluster_A),
                         FUN = mean)

centroid_values <- as.matrix(centroids_A[, -1])

# Assign B records to closest centroid
B_assigned <- apply(B, 1, function(record) {

  distances <- apply(centroid_values, 1, function(center) {
    sqrt(sum((as.numeric(record) - as.numeric(center))^2))
  })

  which.min(distances)
})

B_assigned <- as.integer(B_assigned)

# Full-data clusters
full_clusters <- cutree(as.hclust(hc_ward), k = 4)
full_clusters <- as.integer(full_clusters)

B_full_clusters <- full_clusters[-A_index]

# Compare assignments
stability_table <- table(
  B_assigned,
  B_full_clusters
)

stability_table
```

```
##            B_full_clusters
## B_assigned  1  2  3  4
##          1  0  0  7  0
##          2  3  3  0 10
##          3  0  8  0  0
##          4  0  0  1  4
```

To assess stability, the data were partitioned into two subsets. Ward's hierarchical clustering was applied to partition A, and cluster centroids were computed. Observations in partition B were assigned to the nearest centroid and compared with the cluster assignments obtained from the full dataset. The comparison table indicates that several groups of cereals were assigned to similar clusters across the two solutions, although some differences in cluster membership occurred. This suggests that the clustering solution exhibits a moderate level of stability.

Part 4

The data should not necessarily be normalized if the goal is to identify a cluster of healthy cereals. Normalization gives each variable equal importance in the clustering process, but some variables are much more relevant to health than others. Variables such as calories, sugar, fat, sodium, fiber, and protein should receive greater emphasis because they directly reflect nutritional quality. Variables such as shelf location, consumer ratings, or display characteristics may be less relevant and could be excluded from the clustering analysis. Therefore, rather than normalizing and equally weighting all variables, the clustering should focus primarily on the nutritional attributes that define a healthy cereal.